8.4 Data Displays

Algebra I 8.4 · Student sheet

Name _____ Period ____ Date _____

I can…

- tell whether data are qualitative or quantitative.
- choose an appropriate display for a data set and explain why.
- read and interpret common data displays.

Vocabulary · fill in the missing word

Qualitative (categorical) data — data that are labels or _____, such as colors, names, or zip codes

Quantitative data — numbers that are _____ or counted, such as heights or test scores

Bar graph — compares the counts of _____; the bars don't touch

Circle graph — shows parts of a _____; each section's angle is its share of 360°

Histogram — shows how many _____ values fall in each interval; the bars touch

Line graph — shows how data _____ over time

Box-and-whisker plot — shows the _____ summary and spread of numerical data

Part 1 · Sort the data

Write each description in the correct column: favorite sport · hours of sleep · eye color · number of siblings · zip code · height · brand of phone · temperature each hour · shoe size · jersey number · test score · pizza topping

Qualitative (labels or categories)	Quantitative (measured or counted)

Best display (bar graph, circle graph, histogram, line graph, or box-and-whisker plot) and one reason:

Data	Display	Reason
a. votes for each of 4 lunch options		
b. a puppy's weight each week for 10 weeks		
c. the share of a budget spent on rent, food, and fun		
d. 30 test scores grouped by tens		

Part 2 · Notes: fill in the blanks

Qualitative data are _____; **quantitative** data are numbers that are _____ or _____.

Categories with counts → _____ Parts of a whole → _____

Numbers in intervals → _____ Change over time → _____

In a circle graph, a section's angle = its share · _____.

Part 3 · Examples (work them with the class)

Example 1

Tell whether the data are qualitative or quantitative.

a. favorite colors of students b. heights of plants

c. zip codes of customers d. numbers of pets owned

Example 2

Which display best represents each data set?

a. a city's monthly rainfall over one year

b. the percent of a budget spent on each type of expense

c. the number of students choosing each lunch option

d. the median and spread of the prices of 40 laptops

Example 3

The table shows the favorite seasons of 30 students.

a. What percent of the students chose fall?

b. In a circle graph, what is the central angle of the fall section?

c. Name two good displays for these data.

Season	Spring	Summer	Fall	Winter
Students	9	12	6	3

Example 4

You want to show how many of 24 students scored 60–69, 70–79, 80–89, and 90–100 on a test. Which display should you use? Why do its bars touch?

Part 4 · Partner practice

Partner 1

Tell whether the data are qualitative or quantitative: the number of text messages sent in a day, favorite type of music, area codes, weights of backpacks.

Partner 2

Choose the best display.

a. your height each year from age 5 to age 15

b. the votes for each candidate for class president